noc22_cs107

Mock Exam 1 | Course ID: noc22_cs107

Q1. According to the introductory lecture, which of the following best describes the 'software crisis'?

A) A shortage of available hardware to run new software.
B) The high cost of software licenses compared to hardware.
C) A persistent set of problems where projects are late, over budget, and fail to meet requirements.
D) The rapid obsolescence of programming languages.

Source: ilHPCbkZLV4 | Difficulty: Easy

Q2. Barry Boehm's W5HH principle is a framework used by project managers primarily to:

A) conduct performance reviews for team members.
B) comprehensively define and clarify the project scope.
C) select the appropriate software development lifecycle model.
D) negotiate contracts with external vendors.

Source: nmkKAZ0jatM | Difficulty: Easy

Q3. What is the key difference between a software product targeting a 'horizontal market' versus a 'vertical market'?

A) Horizontal market products are custom-built, while vertical market products are off-the-shelf.
B) Horizontal market products are for a wide range of industries, while vertical market products are for a specific industry.
C) Horizontal market products are information systems, while vertical market products are embedded systems.
D) Horizontal market products are more expensive than vertical market products.

Source: nmkKAZ0jatM | Difficulty: Medium

Q4. Which of the following is NOT one of the five core project management process groups as defined by PMBOK?

A) Initiating
B) Planning
C) Designing
D) Closing

Source: A8-PtgugnGE | Difficulty: Easy

Q5. In the classical waterfall model, which phase is stated to consume the maximum effort among all development and post-delivery phases?

A) Requirements Analysis
B) Design
C) Testing
D) Maintenance

Source: yC0A6_nYPYQ | Difficulty: Medium

Q6. What is the primary reason the V-Model is considered more suitable for high-reliability systems compared to the iterative waterfall model?

A) It allows for more frequent customer feedback.
B) It integrates test planning and design activities into every development phase, not just at the end.
C) It completely eliminates the possibility of defects.
D) It is a risk-driven model that uses prototyping.

Source: J1-LpZr9GSQ | Difficulty: Hard

Q7. What is the key characteristic that distinguishes the RAD model from the traditional prototyping model?

A) RAD uses a strict waterfall approach after prototyping.
B) In RAD, the prototype is refined and evolves into the final product, whereas in the prototyping model, it is typically thrown away.
C) RAD is only suitable for large, high-performance systems.
D) The prototyping model does not involve customer feedback.

Source: 0O6fJSkafOo | Difficulty: Medium

Q8. The Spiral Model is considered a 'meta-model' because:

A) it is the most recently developed model.
B) it is exclusively used for meta-data projects.
C) it can incorporate characteristics of waterfall, prototyping, and iterative models within its loops.
D) it guarantees zero risk in project development.

Source: sQyR78dlSzc | Difficulty: Hard

Q9. In Scrum, what is the primary responsibility of the Product Owner?

A) To facilitate the daily scrum meeting and remove impediments.
B) To write the code for the features in the sprint backlog.
C) To represent the customer's interests, define features, and prioritize the Product Backlog.
D) To manage the technical architecture of the system.

Source: 0JYkgfgKn8I | Difficulty: Easy

Q10. During a Daily Scrum meeting, which of the following questions is NOT one of the three key questions each team member answers?

A) What did I accomplish yesterday?
B) What is my plan for today?
C) How can we solve the technical problem I am facing?
D) Are there any obstacles in my way?

Source: 0JYkgfgKn8I | Difficulty: Medium

Q11. What is the primary purpose of a 'business case' or feasibility study?

A) To create a detailed technical design of the software.
B) To justify the project by demonstrating its financial and technical viability.
C) To hire the development team for the project.
D) To write the user manual for the final product.

Source: 8RuJfk1Xj2U | Difficulty: Easy

Q12. Which project evaluation technique calculates the time required for a project's cumulative cash inflows to equal its initial investment but ignores all profitability after that point?

A) Net Present Value (NPV)
B) Return on Investment (ROI)
C) Payback Period
D) Net Profit

Source: 8UlgpszPK6w | Difficulty: Medium

Q13. A project has an initial investment of $100,000, followed by five annual cash flows of $10,000, $10,000, $10,000, $20,000, and $100,000. According to the lecture's calculation using a 10% discount rate, what is the Net Present Value (NPV)?

A) $50,000

B) $618
C) $10,000
D) -$1,250
Source: 8UlgpszPK6w | Difficulty: Hard

Q14. In the context of program management, what is a 'program'?

A) A single, standalone software application.
B) A group of unrelated projects managed by the same person.
C) A group of related projects managed in a coordinated way to achieve benefits not possible if managed independently.
D) The source code written for a project.
Source: 5W5xe-CJZTQ | Difficulty: Easy

Q15. What is the primary purpose of the Software Project Management Plan (SPMP) document?

A) It is the contract signed with the customer.
B) It is a marketing document for the final product.
C) It is the user manual for the software.
D) It formally records all project plans, estimates, schedules, and risk management strategies.
Source: 0mpuhlabOWE | Difficulty: Medium

Q16. According to the lecture, what is a major shortcoming of using Source Lines of Code (SLOC) as a size estimation metric?

A) It is too difficult to count.
B) It penalizes the use of high-level languages and concise coding styles.
C) It can only be used for object-oriented programming.
D) It is language-independent.
Source: wqyJF-Gbdp0 | Difficulty: Medium

Q17. The Delphi estimation technique is designed to mitigate individual biases and groupthink by:

A) requiring all experts to agree on a single estimate in a face-to-face meeting.
B) using a mathematical formula based on historical data.
C) using multiple rounds of anonymous estimation and feedback facilitated by a coordinator.
D) averaging the estimates of the most senior and most junior team members.
Source: wqyJF-Gbdp0 | Difficulty: Hard

Q18. In PERT/CPM, what does 'Slack' or 'Float' represent?

A) The total duration of the project.
B) The amount of time a non-critical activity can be delayed without affecting the project completion date.
C) The cost overrun on a specific task.
D) The number of resources assigned to a task.
Source: -gqtKDlYSJc | Difficulty: Easy

Q19. During the backward pass in a PERT/CPM calculation, if an activity is a 'burst point' with multiple successors, how is its Latest Finish (LF) time determined?

A) It is the maximum of the Latest Start (LS) times of all its successors.
B) It is the sum of the Latest Start (LS) times of all its successors.
C) It is the minimum of the Latest Start (LS) times of all its successors.
D) It is equal to the project's overall earliest finish time.
Source: 8w2gm0AO-KY | Difficulty: Hard

Q20. What is the primary goal of 'Project Crashing'?

A) To intentionally fail a project to recover insurance costs.
B) To reduce the overall project duration by accelerating specific activities, usually at an increased cost.
C) To remove non-essential features from the project scope.
D) To re-assign all team members to different projects.
Source: ikOTZBPaklE | Difficulty: Medium

Q21. According to Herzberg's Two-Factor Theory, which of the following is a 'Motivator' (or satisfier)?

A) Company policy
B) Salary
C) Working conditions
D) Recognition of effort and performance
Source: TQWJYqOzs0g | Difficulty: Medium

Q22. What is the primary disadvantage of a 'Chief Programmer Team' structure?

A) Decision-making is too slow due to extensive discussions.
B) It is only suitable for very large and complex projects.
C) The project is highly dependent on a single individual, creating a single point of failure.
D) It leads to poor code quality due to a lack of expertise.
Source: wjd0YMixlJA | Difficulty: Hard

Q23. What is the primary purpose of a resource histogram in project management?

A) To track the project budget over time.
B) To visualize the demand for a specific resource over the project's duration.
C) To list the dependencies between project activities.
D) To document the project's final deliverables.
Source: ZcO5UNzMtz8 | Difficulty: Easy

Q24. In Earned Value Analysis (EVA), what does a Cost Performance Index (CPI) of less than 1 indicate?

A) The project is under budget.
B) The project is ahead of schedule.
C) The project is over budget.
D) The project is exactly on schedule and on budget.
Source: I81fOjyzb0E | Difficulty: Medium

Q25. What is the primary function of a Change Control Board (CCB)?

A) To write the initial project requirements.
B) To test the final software product.
C) To review and approve proposed changes to controlled work products.
D) To manage the project's budget.
Source: uITAo7fbm3M | Difficulty: Easy

Q26. What is the key difference between a 'Fixed Price' contract and a 'Time and Material' contract?

A) Fixed price contracts are only for hardware, while time and material are for software.
B) In a fixed price contract, the customer has budget certainty, while in a time and material contract, the customer bears the financial risk of overruns.
C) Fixed price contracts are more flexible for changing requirements.
D) Time and material contracts motivate suppliers to be more cost-effective.
Source: olrW1tqGnsM | Difficulty: Medium

Q27. What is the primary purpose of a post-implementation project review, also known as a 'post-mortem analysis'?

A) To assign blame for project failures.
B) To calculate the final financial profit or loss of the project.
C) To identify successes, failures, and lessons learned to improve future projects.
D) To release the final software version to the public.
Source: TfvonvpVeTE | Difficulty: Easy

Q28. What is a fundamental distinction between ISO 9001 certification and SEI CMM assessment?

A) ISO 9001 is for software only, while SEI CMM is for all industries.
B) ISO 9001 provides a step-by-step improvement path, while SEI CMM is a go/no-go certification.
C) ISO 9001 is a generic quality assurance standard, while SEI CMM is a software-specific framework providing a gradual path for process maturity improvement.
D) Only SEI CMM requires configuration management.
Source: k6I04YmQWl8 | Difficulty: Hard

Q29. In the context of software reliability, what does the '90-10 rule' state?

A) 90% of bugs are found by 10% of testers.
B) A project is 90% complete with 10% of the work remaining.
C) Approximately 90% of a program's execution time is spent in only 10% of its code.
D) 90% of development effort is spent on coding, and 10% on testing.
Source: bfx1u27hvvU | Difficulty: Medium

Q30. According to Putnam's model, if a project's development time (td) is compressed by half, the total effort (K) required increases by a factor of:

A) 2
B) 4
C) 8
D) 16
Source: _KdYrzzMuBo | Difficulty: Hard

Answer Key

1: C	2: B	3: B	4: C	5: D
6: B	7: B	8: C	9: C	10: C
11: B	12: C	13: B	14: C	15: D
16: B	17: C	18: B	19: C	20: B
21: D	22: C	23: B	24: C	25: C
26: B	27: C	28: C	29: C	30: D